

RESEARCH ARTICLE

Novel class identification: A human-in-the-loop approach for efficient species recognition in bioacoustic data

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Abstract

1. Passive acoustic monitoring (PAM) is widely used to collect large-scale wildlife data encompassing many species, but efficiently identifying all species—particularly rare and potentially endangered ones—without exhaustive manual examination of the entire dataset remains a major challenge.
2. We formally introduce the machine learning task of novel class identification (NCI), which aims to identify as many classes (e.g. species) as possible while examining as few samples as necessary, and we propose a comprehensive evaluation framework for both overall and class-presence-specific performance. NCI uses human-in-the-loop querying strategies based on the active learning paradigm, while its objective of efficiently identifying classes aligns with open-world learning assumptions and is closely related to the goals of novel class discovery. Drawing on insights from the literature, we identify four promising query method families: embedding-based methods, self-supervised learning, similarity learning and audio-language models.
3. To explore the potential of these directions, initial experiments evaluate 44 query method configurations from the identified query method families across five multi-label PAM datasets. The results demonstrate that the proposed evaluation framework effectively captures both overall and class-presence-specific performance, enabling robust comparison of different methods. These experiments further indicate that audio-language models achieve the highest performance, providing a foundation for future research.
4. Our findings highlight the potential of deep learning methods for rapid species identification, supporting faster and more informed conservation decisions from large PAM datasets.

KEYWORDS

audio-language models, human-in-the-loop, novel class identification, passive acoustic monitoring, species identification

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1 | INTRODUCTION

Anthropogenic activities—including pollution, habitat loss, hunting, introduction of invasive species, overexploitation, climate change and natural disasters—have made humanity the primary driver of species extinction (Singh et al., 2021; Ruckelshaus et al., 2020), with rates 100–1000 times higher than pre-industrial levels (De Vos et al., 2015; Pimm et al., 2014). To mitigate this threat, conservation initiatives act when endangered species are identified, ranging from broad legislation, such as North America's Endangered Species Act (US Fish and Wildlife Service, 1973), to site-specific measures, including habitat protection and seasonal construction restrictions in Tongzhou Bay, China, safeguarding shorebirds and other wildlife (Muller et al., 2020). Fast and accurate species identification is critical for these efforts, as many stakeholders view biodiversity protection as a constraint and rely on rapid assessments to inform conservation measures (Bennett et al., 2024).

Efficient and timely species identification is therefore essential, and passive acoustic monitoring (PAM) has emerged as a key tool for cost-effective, large-scale wildlife assessment. PAM systems continuously record environmental soundscapes, providing insights into species richness, animal behaviour and ecosystem health (Gibb et al., 2019; Ross et al., 2023; Sugai et al., 2019; Sugai & Llusia, 2019). However, despite enabling scalable data acquisition, manual analysis of these vast recordings is infeasible, and automatic analysis remains challenging due to undirected recording setups, overlapping species calls, distant target sounds and environmental noise such as wind and rain (Browning et al., 2017; Enari et al., 2026; Oswald et al., 2022; Sethi et al., 2023).

To address this challenge, XPrize Rainforest, 'a global 5-year, \$10 million competition that convenes innovators and experts across disciplines [...] and challenges them to use novel technologies to expedite the monitoring of tropical biodiversity' (<https://www.xprize.org/prizes/rainforest>), requires teams to survey 100 hectares of tropical rainforest within 24 h, with no human entry allowed. The goal is to generate the most extensive list of species possible by analysing the data within an additional 48 h. Scalable and non-invasive, PAM is a natural choice for audio data acquisition in this challenge, presenting the problem of identifying as many species as possible from the recordings, given the limited time and specialized expertise required (Kath et al., 2024a). We introduce this task as novel class identification (NCI), which uses human-in-the-loop query methods to iteratively select samples (e.g. audio snippets) from a large unlabelled dataset, aiming to identify as many previously unseen classes (e.g. species or call types) as possible while examining the fewest samples necessary. The remainder of the introduction situates the NCI task within existing literature, emphasizing its connections and distinctions relative to related machine learning paradigms.

1.1 | Active learning

Active learning uses human-in-the-loop query methods which, given a large unlabelled dataset, a small labelled dataset, and

limited annotation capacity, iteratively select unlabelled samples for expert annotation, followed by dataset updates. Active learning query methods aim to select samples that maximize model performance for predefined classes, enhancing the informativeness of the labelled dataset and enabling faster convergence with fewer annotations (Settles, 2009). These methods can be categorized into three types: Prediction-based methods use model performance to select the next query, for example uncertainty sampling prioritizes samples near the decision boundary. Data-based methods select samples based on the internal data structure, for example diversity sampling prioritizes samples that represent the entire input space. Model-based methods focus on changes in the model, for example prioritizing samples with high model influence, regardless of their label (Tharwat & Schenck, 2023).

In bioacoustics, active learning has shown strong performance across various dataset types. Wang et al. (2022) use a synthetic dataset, created by recombining environmental sounds with urban soundscape backgrounds, to explore active learning. Qian et al. (2017) apply active learning to improve the data efficiency of bird species classifiers using a museum sound collection, likely composed of focal, single-label recordings. Allen et al. (2021) use active learning to detect humpback whale songs in a large single-label PAM dataset of 187,000 h. van Osta et al. (2023) and Kath et al. (2024b) demonstrate that active learning is particularly effective for rare species. Kath et al. (2024a) demonstrate the efficiency of various basic active learning query methods compared to random sampling on one synthetic and two real-world, manually annotated multi-label PAM datasets. Rauch et al. (2024) extend these findings through initial experiments with state-of-the-art query methods on a 7-h real-world PAM dataset of bird vocalizations.

NCI uses the same human-in-the-loop process as active learning. However, the objectives and underlying assumptions differ significantly, as active learning focuses on optimizing a classifier in a *closed-world* setting with a fixed label space (Settles, 2009), where the set of classes is assumed to be known and unchanged during training.

1.2 | Open-world learning

A more realistic scenario is the *open-world* assumption (Drummond & Shearer, 2006), which permits distribution shifts between training and test data. Given an in-distribution dataset $\mathcal{D}_{\mathcal{I}\mathcal{D}}$ with input (sensor) space $\mathcal{X}$ and output (label) space $\mathcal{Y}$, its data distribution is represented by $P(X, Y)$ over the joint space $\mathcal{X} \times \mathcal{Y}$. An out-of-distribution sample x_{OOD} can exhibit two types of distribution shift: A covariate shift alters the input distribution $P(X)$ while keeping $\mathcal{Y}$ fixed (Wang & Deng, 2018); this occurs when x_{OOD} originates from a different domain, for example in bioacoustics, due to variations in background soundscape, species sub-population, event occurrence frequency or microphone type (Stowell, 2022). A semantic shift directly alters the output distribution $P(Y)$, which in turn indirectly affects the input distribution $P(X)$ (Hendrycks & Gimpel, 2017); this occurs when

x_{OOD} belongs to a class not present in $\mathcal{Y}$, for example in bioacoustics, when recordings contain acoustic classes (e.g. species or call types) not present in the training set.

A variety of tasks focussing on the detection of out-of-distribution data have been derived from the open-world assumption, with the most frequently used ones summarized in Table 1. Anomaly detection (Chandola et al., 2009) focusses on identifying samples that exhibit covariate or semantic shifts from the predefined normality during testing. Novelty detection (Markou & Singh, 2003; Miljković, 2010) is often used interchangeably with semantic anomaly detection. The subtle differences are that anomaly detection identifies erroneous samples and may utilize out-of-distribution data during training, while novelty detection treats identified samples as new learning resources and is typically fully unsupervised (Yang et al., 2024). Open set recognition detects semantic shifts to prevent overconfident misclassification of unknown classes as known ones (Scheirer et al., 2013), sharing goals with multi-class novelty detection while additionally requiring accurate classification of known classes. Out-of-distribution detection (Hendrycks & Gimpel, 2017) is closely related to open set recognition but differs by using separate datasets for in- and out-of-distribution data, allowing out-of-distribution data during training and addressing a broader range of tasks, such as multi-label classification. Outlier detection (Wang et al., 2019) aims to identify covariate or semantic shifts within an entire dataset by analysing all observations jointly, and therefore differs from other methods by not adhering to the classical train-test split.

Bioacoustic data recorded using PAM inherently involves covariate and semantic distribution shifts; however, most research assumes closed-world scenarios (Ghani et al., 2023; Kath et al., 2024a). Elaborations considering covariate shifts evaluate systems not only on test sets from the same data source but also on datasets with varying location, signal-to-noise ratio or season (Allen et al., 2021; Khalighifar et al., 2021). Best et al. (2020) explore the automatic

transferability of models under covariate shifts, using convolutional neural networks and treating features from the first convolutional layers as new dimensions to detect orca vocalizations. To improve work considering semantic shifts, Stowell et al. (2019) designed the open-set recognition challenge 'bird audio detection' with the goal of predicting the presence of any bird in a short audio recording. Cramer et al. (2020) introduce a hierarchical classification model for open-set recognition that classifies coarse taxa (e.g. animal order) even when the fine taxa (e.g. species) is novel. Bjerger et al. (2023) develop a hierarchical classification framework combining multitask learning and anomaly detection to categorize insects across taxonomic ranks and detect novel species at higher levels of taxonomy. Zhu et al. (2022) propose a class-incremental learning approach that efficiently integrates previously unseen species as new classes without retraining the entire model. Treating species not included in the training set as outliers, Ntalampiras and Potamitis (2021) train a variational autoencoder, an algorithm that compresses and reconstructs data, on the labelled dataset; as the model is trained to reconstruct known data, high reconstruction error is assumed to indicate a semantic shift.

NCI follows the open-world assumption and aims to detect out-of-distribution data with semantic shifts, but it differs from open-world learning methods in several ways. Key differences include the iterative changes of the labelled and unlabelled sets in NCI, resembling active learning, and that open-world learning's semantic shift detection does not distinguish whether out-of-distribution samples belong to the same or different novel classes, limiting its efficiency for class identification.

1.3 | Novel class discovery

Refining semantic shift detection in open-world settings, the novel class discovery task leverages knowledge from labelled in-distribution data $\mathcal{D}_L$ to identify classes in unlabelled out-of-distribution data $\mathcal{D}_U$, typically via unsupervised methods. Most setups use disjoint classes for $\mathcal{D}_L$ and $\mathcal{D}_U$ and a predefined number of classes in both datasets (Troisemaine et al., 2023).

Similarity-based methods (Hsu et al., 2018) use pairs of instances from $\mathcal{D}_L$ to train a binary classifier that learns if a pair belongs to the same class. Applying the classifier to all unique pairs of $\mathcal{D}_U$ produces the prediction matrix $\tilde{Y}_U \in \mathbb{R}^{M \times M}$ ($M = |\mathcal{D}_U|$), which is used as a target to train a deep clustering network on $\mathcal{D}_U$ for the final class prediction. Embedding-based methods (Han et al., 2019) use $\mathcal{D}_L$ to train a function that produces meaningful representations, which, assuming shared high-level features between $\mathcal{D}_L$ and $\mathcal{D}_U$, are used to embed $\mathcal{D}_U$ for subsequent clustering. The most prominent approach to learning the embedding function is transfer learning (Pan & Yang, 2010), where a neural classifier is trained on $\mathcal{D}_L$ and the classification layer is removed. End-to-end methods (Han et al., 2022) exploit both $\mathcal{D}_L$ and $\mathcal{D}_U$ simultaneously, addressing the limitations of similarity-based and latent-space-based methods that are negatively

TABLE 1 Comparison of the open-world tasks anomaly detection (AD), novelty detection (ND), open set recognition (OSR), out-of-distribution detection (OODD) and outlier detection (OD). The comparison is based on the detected distribution shift (covariant, semantic); whether the in-distribution (ID) label set is single-class (SC) or multi-class (MC); if, in addition to detecting out-of-distribution (OOD) samples, the ID classes must also be correctly classified; and whether OOD data is included in the training dataset.

Task	Shift		$\mathcal{Y}_{\text{ID}}$		ID-classifier	$\mathcal{D}_U$
	Cov.	Sem.	SC	MC		
AD	✓	✓	✓	✓		(✓)
ND		✓	✓	✓		
OSR		✓		✓	✓	
OODD		✓		✓	✓	(✓)
OD	✓	✓	✓	✓		✓

impacted when relevant high-level features are not fully shared between the classes of $\mathcal{D}_L$ and $\mathcal{D}_U$. The model architecture consists of a shared encoder with two subsequent networks: a classification network trained on $\mathcal{D}_L$ and a clustering network trained on $\mathcal{D}_U$ using generated pseudo-labels.

In bioacoustics, novel class discovery typically relies on transforming audio recordings into feature representations that enable comparison and clustering. Guerrero et al. (2023) generate embeddings from noise-reduced recordings using linearly spaced cepstral coefficients and frequency features, which are then clustered and evaluated across birds, amphibians, mammals and insects. Most approaches leverage pre-trained models to generate embeddings. Calonge et al. (2024) employ AVES (Hagiwara, 2023), a self-supervised transformer model for encoding animal vocalizations, together with a convolutional autoencoder to generate embeddings. These embeddings are then reduced in dimensionality using principal component analysis and clustered with HDBSCAN (Campello et al., 2013) to organize biophony, geophony, anthrophony and their subcategories. Allen-Ankins et al. (2025) use BirdNET (Kahl et al., 2021), a neural network trained on multi-class bird vocalizations, to generate embeddings and measure the similarity of an unlabeled audio sample to a target sound via Euclidean distance. Sampling audio close to the target helps identify species with similar call types. Kath et al. (2024a) and Kather et al. (2025) further demonstrate that BirdNET embeddings form visually coherent clusters corresponding to previously unseen species.

Novel class discovery and NCI both aim to detect new classes in an unlabelled dataset. However, while NCI focusses on efficiently identifying new classes, novel class discovery aims to properly cluster $\mathcal{D}_U$. This results in three key differences: (1) Instead of using fixed $\mathcal{D}_L$ and $\mathcal{D}_U$ with disjoint classes, these sets are updated iteratively in NCI. (2) Imbalanced datasets are handled differently, as novel class discovery is primarily evaluated by clustering accuracy (Yang et al., 2010), which measures the proportion of samples in each cluster that belong to the same class, giving equal importance to each sample. In contrast, NCI assigns equal importance to each class, regardless of class balance. (3) NCI assumes an unknown number of classes. While this is typically not the case for novel class discovery, methods that assume an unknown number of classes in $\mathcal{D}_U$ usually either initialize a large number of clusters (e.g. 100 in Hsu et al., 2018), allowing the clustering network to use only the necessary clusters, or determine the number of clusters that maximizes clustering accuracy (Vaze et al., 2022).

1.4 | Transfer learning

Building on pre-trained models in novel class discovery, transfer learning leverages large labelled datasets to improve feature representations and model performance on related tasks, including classification, few-shot learning and embedding generation (Pan & Yang, 2010). Different layers of deep neural networks encode

progressively higher levels of abstraction, from raw input in the first layer to semantic class in the last, enabling large models trained on extensive source datasets to transfer to target tasks that share abstract features (Azizpour et al., 2015). The classification layer is adapted to the target data, while the remaining layers can either be kept frozen or fine-tuned from their pre-trained weights (Azizpour et al., 2015; Yosinski et al., 2014).

As bioacoustic data labelling is time- and cost-intensive, transfer learning facilitates the interpretation of new datasets. Nolasco et al. (2023) use pre-trained models to detect novel animal sounds from as few as five labelled examples. Swaminathan et al. (2024) fine-tune wav2vec for multi-label bird species classification, demonstrating that transfer learning effectively captures diverse acoustic features for detecting multiple species within real-world recordings. Kath et al. (2024a) and Ghani et al. (2023) demonstrate that BirdNET (Kahl et al., 2021), a neural network trained on multi-class bird vocalizations, excels at generating embeddings suitable for both multi-class and multi-label classification of PAM data. Building on this, Weldy et al. (2026) employ BirdNET embeddings to train a classifier using only two labelled samples per class. Extending this line of work, Kather et al. (2025) demonstrate that BirdNET embeddings also outperform alternative representations in latent-space clustering for novel class discovery. Additional experiments further indicate that fine-tuning BirdNET on the target species improves the quality of its embeddings (Kath et al., 2025a, 2025b). For further reading, a comprehensive overview of bioacoustic foundation models is provided by Schwinger et al. (2026), and a detailed methodology for creating generalizable bioacoustic encoders is presented by Miron et al. (2026).

Transfer learning is widely used in bioacoustics, and models such as BirdNET generate meaningful embeddings that form coherent clusters of previously unseen species, making this approach promising for NCI.

1.5 | Audio-language models

Transfer learning laid the foundation for transformer models (Vaswani et al., 2017), a versatile architecture that learns both general and domain-specific representations across diverse modalities, enabling zero-shot transfer—applying knowledge to novel tasks without explicit retraining—through a unified representational interface. Audio-language models, composed of audio and text encoders jointly trained on audio-text data, are capable of processing and reasoning about sounds (Su et al., 2025).

This is particularly beneficial for niche domains such as bioacoustics, as it reduces reliance on manual annotations and predefined label spaces (Miao et al., 2025). Robinson et al. (2024) introduce BioLingual, the first audio-language model fine-tuned on bioacoustic data via contrastive learning, achieving 68.9% zero-shot classification accuracy on 29,134 samples across 1143 species, and 63% mean average precision@1 in text-to-audio retrieval. Miao et al. (2025)

train CLAP (Elizalde et al., 2023), an audio-language model with contrastive learning, on non-bioacoustic audio-text pairs, demonstrating strong zero-shot transfer for bioacoustic classification and species detection across various datasets, but lacking accuracy in fine-grained species-level categorization. Robinson et al. (2025) extend this line with NatureLM-Audio, the first audio-language foundation model tailored to bioacoustics, achieving promising results in classification and novel class discovery tasks. NatureLM-Audio is based on BEATs (Chen et al., 2023) for audio encoding and Llama 3.1-8b (Dubey et al., 2024) for text processing, with the Llama component fine-tuned specifically on animal vocalizations.

Audio-language models are relatively new to the domain of bioacoustics, and only limited research has explored their potential so far. Nevertheless, their ability to transfer knowledge across tasks and perform zero-shot learning makes them a promising direction for NCI.

2 | MATERIALS AND METHODS

This section formally defines the novel class identification (NCI) task, outlines the framework for qualitative and quantitative evaluation, summarizes promising query method families from the literature and details the datasets, query methods and set-up used for an initial experimental study.

2.1 | Formal task definition

Figure 1 illustrates the NCI process from a data perspective. The dataset $\mathcal{D}$ contains C classes, where C is unknown. At iteration t , $\mathcal{D}_U^{t-1}$ and $\mathcal{D}_L^{t-1}$ denote the previous unlabelled and labelled dataset, with $\mathcal{D}_L^{t-1}$ containing C^{t-1} classes. NCI uses a human-in-the-loop querying strategy, inspired by active learning (Kath et al., 2026), to iteratively select samples for expert annotation: (1) A query method $QM(\mathcal{D}_U^{t-1}, \mathcal{D}_L^{t-1}, t)$ selects n^t samples from $\mathcal{D}_U^{t-1}$ to form the unlabelled query $\mathcal{Q}_U^t$, where n^t is the query size for iteration t . (2) $\mathcal{Q}_U^t$ is labelled by human experts and becomes $\mathcal{Q}_L^t$. (3) Datasets are updated: $\mathcal{D}_U^t = \mathcal{D}_U^{t-1} \setminus \mathcal{Q}_U^t$ and $\mathcal{D}_L^t = \mathcal{D}_L^{t-1} \cup \mathcal{Q}_L^t$, with $\mathcal{D}_L^t$ containing C^t classes. The next iteration begins unless $n^t = 0$. The task of NCI is to determine a query method that, at each iteration, maximizes the

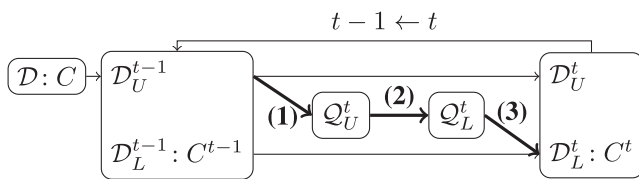


FIGURE 1 Novel class identification: The dataset $\mathcal{D}$ with C classes is initially unlabelled. (1) A query method uses the current unlabelled dataset $\mathcal{D}_U^{t-1}$ and labelled dataset $\mathcal{D}_L^{t-1}$ with C^{t-1} classes to query samples ($\mathcal{Q}_U^t$) from $\mathcal{D}_U^{t-1}$. (2) $\mathcal{Q}_U^t$ is labelled ($\mathcal{Q}_L^t$) by a human expert. (3) Update datasets: $\mathcal{D}_U^t = \mathcal{D}_U^{t-1} \setminus \mathcal{Q}_U^t$ and $\mathcal{D}_L^t = \mathcal{D}_L^{t-1} \cup \mathcal{Q}_L^t$ with C^t classes.

number of newly identified classes while minimizing the number of manually assessed samples. Formally, this is defined as

$$QM^*(\mathcal{D}_U^{t-1}, \mathcal{D}_L^{t-1}) = \max_{QM} (\lambda^t (C^t - C^{t-1}) - n^t) \forall t,$$

where λ^t represents the cost-efficiency factor, indicating how many samples are worth investigating per newly identified class. As NCI is introduced using PAM data, we assume $\mathcal{D}$ to be multi-label, where each sample x_i with index i has a true label $y_i \in \mathcal{Y} = \{0, 1\}^C$, and its label at iteration t is given by $y_i^t \in \mathcal{Y}^t = \{0, 1\}^{C^t}$. Therefore, multiple classes can be identified from a single sample, and the predicted presence of an identified class does not preclude the existence of other unidentified classes. Classes correspond to user-defined units of interest, which may vary depending on the task definition; for example, in the context of PAM, classes may refer to species, species-specific call types, or broader soundscape-level acoustic patterns, among others. The class set C may represent predefined sets; for example, it may correspond to a list of endangered species when the objective is to identify their presence within a planned construction area to inform decision-making. Alternatively, C may represent previously unspecified classes, for example in settings aimed at identifying unknown species. It may also comprise both, for example in large-scale biodiversity assessment settings such as XPRIZE Rainforest, where C contains all animal species, as the goal is to produce the most comprehensive list of species present.

2.2 | Evaluation framework and metrics

The primary objective of NCI query methods is to maximize the number of classes identified while minimizing the number of samples required to identify them. Because sparse classes are harder to detect (Bennett et al., 2024), performance on the labelled dataset $\mathcal{D}_L^t$ is evaluated both by the total number of classes identified and by the relationship between class occurrence and identification.

Random sampling is the simplest baseline for sample selection (Kath et al., 2026); thus, all results are reported relative to it. The random sampling baseline for NCI can be analytically derived, as the number of samples required to detect a class with a given probability depends only on its presence in the dataset. The probability that a class c , containing n_p^c positive samples in a dataset of size n_{DS} , is identified within the first n_s^c selected samples using random sampling is given by

$$P_{\text{found}}^{c,rs} \left(n_p^c, n_{DS}, n_s^c \right) = 1 - \prod_{i=0}^{n_s^c-1} \left(1 - \frac{n_p^c}{n_{DS} - i} \right). \quad (1)$$

Assuming that $n_s^c \ll n_{DS}$ and $n_p^c \ll n_{DS}$, Equation (1) can be approximated as

$$P_{\text{found}}^{c,rs} \left(n_p^c, n_{DS}, n_s^c \right) \approx 1 - \left(1 - \frac{n_p^c}{n_{DS}} \right)^{n_s^c}. \quad (2)$$

From this, the expected number of random samples required to identify class c with probability P_{found}^c is

$$n_s^{c,rs} \left(n_p^c, n_{DS}, P_{\text{found}}^c \right) = \frac{\ln(1 - P_{\text{found}}^c)}{\ln(1 - \frac{n_p^c}{n_{DS}})} \quad (3)$$

We propose evaluating query methods by comparing their performance to random sampling, with the identification probability set to 50%. To compute the random sampling baseline analytically, P_{found}^c is set to 0.5 and $n_s^{c,rs}$ is calculated for all classes. For query methods relying on more complex patterns, analytical computation is infeasible; performance is instead evaluated empirically over multiple runs, with $n_s^{c,qm}$ denoting the number of samples required for class c to be identified in at least 50% of runs.

Iterative processes such as NCI lack widely established metrics for quantitative performance evaluation. A key limitation is that performance curves are inherently dataset-dependent, since species vary in acoustic complexity, variability and detectability, which affects the number of samples required for confident identification (Hutschenreiter et al., 2023). Despite this, we propose reporting both qualitative and quantitative results for overall and class-presence-specific performance as follows:

Overall performance can be assessed qualitatively using plots that show, for each query method, the number of identified classes as a function of the number of labelled samples, alongside the random sampling baseline derived from Equation (3). Quantitative evaluation approximates the learning curve and computes a speed-up factor, indicating the fraction of samples a query method requires to reach the same performance as random sampling, originally developed for the iterative active learning process (Kath et al., 2026). Following Kath et al. (2024b), we approximate the learning curve as follows:

$$C_{\text{found}}(n_L^t) = c_{\text{all}} \left(1 - e^{-\frac{n_L^t}{b}} \right), \quad (4)$$

with the asymptote c_{all} set to the total number of classes in the dataset. C_{found} represents the number of identified classes, n_L^t the number of labelled samples, and the parameter b is fitted for each query method using least squares. By fitting b_{qm} for a query method and b_{rs} for random sampling, the speed-up factor $S = \frac{b_{qm}}{b_{rs}}$ quantifies the fraction of samples a query method needs to examine to reach the same number of identified classes as random sampling, with lower values indicating better performance. Specifically, $S = 1$ indicates no improvement over random sampling, $S < 1$ reflects a relative efficiency gain, and $S > 1$ denotes worse performance than random sampling (for proof, see Kath et al. (2026)).

Class-presence-specific performance for a query method can be assessed qualitatively using plots showing the number of samples required to identify each class relative to its presence in the dataset. The random sampling baseline, computed using Equation (3) at $P_{\text{found}}^c = 0.5$, serves as a reference line. Classes above this line require more samples, and those below fewer, to be identified by the query method compared to random sampling. For quantitative evaluation, we introduce the relative identification score (RIS), which measures the average performance gain or loss of a query method relative to random sampling across all classes. RIS ranges from -1 to 1, with the random-sampling RIS lying between 0 and

1 depending on the total query budget. Values below the random-sampling baseline indicate decreased performance, while values above it indicate improved performance. Larger absolute differences from the baseline reflect greater deviation. The metric is approximated as shown in Equation (5), where C denotes the set of classes, n_p^c the number of positive samples for class c , n_{DS} the total number of samples in the dataset, $n_s^{c,qm}$ the number of samples required for class c to be identified in at least 50% of runs, and $n_{s,\text{max}}$ the maximum number of samples evaluated. A detailed definition is provided in Appendix A.

$$RIS(qm) \approx -1 + \frac{2}{|C|} \sum_{c \in C} \left(1 - \frac{n_p^c}{n_{DS}} \right)^{\min(n_s^{c,qm}, n_{s,\text{max}})} \quad (5)$$

Processing time is reported alongside performance, as in iterative human-in-the-loop settings the computation time per iteration directly affects the annotator's workflow. We therefore consider only the processing time incurred during each iteration, excluding one-time embedding or preprocessing steps performed prior to the iterative process. Processing time is reported qualitatively as the per-iteration computation time, normalized relative to random sampling, and quantitatively as the mean per-iteration time.

2.3 | Promising families of query methods

Building on the related work, we distil four research directions that can be adapted for selecting samples from an unlabelled dataset containing classes not present in the current labelled dataset: embedding-based methods, self-supervised learning methods, similarity learning methods, and audio-language model methods.

2.3.1 | Embedding-based methods

Embedding PAM data using pre-trained models has been shown to produce effective class separation (e.g. Bravo Sanchez et al., 2024; Ghani et al., 2023). For classification tasks, using frozen embeddings from the penultimate BirdNet layer yields strong performance (Kath et al., 2024a; McGinn et al., 2023), while fine-tuning the model can further improve results (Kath et al., 2025a; Márquez-Rodríguez et al., 2025; Schiavo et al., 2025).

For NCI, embeddings can support query methods based on either diversity or uncertainty sampling adapted from active learning. Adapted diversity methods operate in the embedding space, based on the idea that species form clusters, so sampling from different regions increases the chance of identifying new classes. Dimensionality reduction has shown to improve clustering quality for PAM data (Kath et al., 2024a; Kather et al., 2025). Adapted uncertainty methods operate on the classification output, aiming to select samples likely to contain a new class. Initial experiments train two classifiers on $D_L^t: C_L^t$, one detecting the presence of any species and another detecting already identified species, and apply them to

D_{ij}^t to prioritize samples likely to contain any species but unlikely to contain already identified species (Kath et al., 2024a).

2.3.2 | Self-supervised learning methods

Self-supervised learning methods leverage inherent data structures, which is particularly advantageous for NCI, where no labelled data are available initially.

Approaches such as contrastive learning and masked prediction models (e.g. wav2vec 2.0, Baevski et al., 2020 and AVES Hagiwara, 2023) learn generalizable audio representations without annotations. Variational autoencoders (Kingma & Welling, 2014; Rezende et al., 2014) encode inputs into a probabilistic latent representation and reconstruct them via a decoder. They can identify novel species through high reconstruction error, defined as the difference between input and reconstruction (Ntalampiras & Potamitis, 2021). This approach can be extended to semi-supervised learning by adding a shallow classification head to the encoder output (Kath et al., 2023), shaping the latent space for improved class separation.

2.3.3 | Similarity learning methods

Similarity learning methods aim to predict whether two samples belong to the same class (Hsu et al., 2018), making them well-suited for NCI. These approaches leverage relationships between sample pairs to structure the embedding space such that samples from the same class are closer together, while samples from different classes are farther apart. Related approaches such as contrastive learning use positive and negative pairs to maximize agreement and separation, and have been shown to provide generalizable audio embeddings in bioacoustics (Miao et al., 2025; Robinson et al., 2024). Different similarity definitions can be used: weak similarity, where samples share at least one class, and strong similarity, where class sets are identical; samples with no class are considered similar in both cases (Zhang & Zhou, 2014). By evaluating pairwise similarities between labelled and unlabelled samples,

these methods can identify samples that are most distinct from known classes, increasing the likelihood of identifying new classes.

2.3.4 | Audio-language model methods

Leveraging general knowledge has proven effective across bioacoustic tasks, with audio-language models such as NatureLM-Audio performing particularly well on diverse benchmarks (Robinson et al., 2025). Using audio-prompt pairs, the model can be prompted to directly output the present classes or, following prior work on hierarchical classification in bioacoustics (e.g. Cramer et al., 2020; Stowell et al., 2019), predict taxonomic ranks from domain to class. In the latter case, only reliably identified levels are reported, enabling potential recognition of novel classes via higher-level taxa.

2.4 | Experimental study

This section presents an initial experimental study of NCI using query methods based on the method families derived in Section 2.3, aiming to validate the framework described in Section 2.2 and inform future research on NCI.

2.4.1 | Datasets

We use five manually annotated, multi-label real-world PAM datasets (Table 2), all of which exhibit strong class imbalance, as illustrated in Figure 2. For pre-processing, the datasets are standardized by generating 2-s audio snippets and resampling to 48 kHz.

2.4.2 | Query methods

Baseline

We use random sampling as a naive baseline. To assess the contribution of model pre-training, we additionally implement a random feature projection baseline, denoted as Random CNN, following prior

TABLE 2 Overview of manually annotated, multi-label passive acoustic monitoring datasets used in this study.

Dataset (ref.)	Recording location	Size (h)	# species	Annotated taxa
AADD (Weldy et al., 2024)	West Coast, USA	11	58	Birds, Mammals, Amphibians
AnuraSet (Cañas et al., 2023)	Brazil	26	42	Frogs, Toads
BirdSet PER (Hopping et al., 2022)	Peru	21	130	Birds
ECOSoundSet ^a (Funosas et al., 2026)	Europe	27	200	Orthopterans, Cicadas
WABAD ^a (Pérez-Granados et al., 2026)	Worldwide	70	1116	Birds

^aExcluding weakly annotated samples.

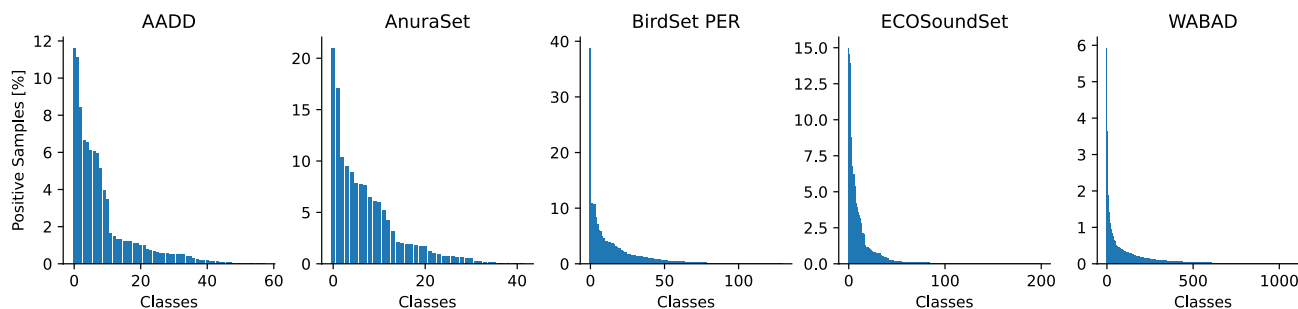


FIGURE 2 Class-wise occurrence frequency for each dataset used.

work (Pons & Serra, 2019). Specifically, we use a randomly initialized two-layer convolutional neural network (64 and 16 filters) to generate 16-dimensional embeddings. Samples are then selected within this embedding space using the Core-Set method described below.

Active learning methods

Within embedding-based methods, we evaluate adapted active learning methods using audio embeddings generated from the penultimate layer of BirdNET, which outputs a 1024-dimensional vector. For all methods, embeddings from the frozen BirdNET model are compared to those obtained after fine-tuning the entire BirdNET model on the current labelled dataset D_L^{t-1} . Fine-tuning is performed by adding a classification layer with one node per class and retraining the full model; the classification layer is subsequently removed, and 1024-dimensional embeddings are extracted from the updated network. We test three adapted diversity-based methods, evaluating embeddings reduced via principal component analysis from the original 1024 dimensions to 2, 16 and 128, as well as retaining all 1024 dimensions. These settings span a range from strong compression to no compression and are equidistant on a logarithmic (base-2) scale, enabling an assessment of sensitivity to information loss.

1. Core-Set (Sener & Savarese, 2018) iteratively selects the unlabelled sample that maximizes the minimum Euclidean distance to the current labelled set in the embedding space.
2. HDBSCAN (Campello et al., 2013) identifies clusters of varying shapes and densities while automatically detecting outliers. In each iteration, 80% of the samples are selected uniformly across clusters and 20% from outliers.
3. K-means (McQueen, 1967) partitions data into a predefined number of clusters. Following Kath et al. (2024a), the number of clusters is set to the sum of the number of already identified classes and half the number of samples to be selected. The centroid and a random sample from the unlabelled set are iteratively selected uniformly across clusters.

We test two adapted uncertainty-based methods:

1. Ratio-max (Monarch, 2021) computes a class-wise uncertainty score $U = \frac{1}{0.5 + |y^* - 0.5|} - 1$; $U \in [0, 1]$ from the predicted value

$y^* \in [0, 1]$. Following Kath et al. (2024a), the maximum score across all classes is used as the sample-level uncertainty, and the unlabelled samples with the highest scores are selected.

2. Dual-Classifier (Kath et al., 2024a) uses the current labelled dataset to train one classifier to predict the presence of any species $y_{any}^* \in [0, 1]$ and another to predict the presence of each already identified species $y_{spec}^* \in [0, 1]$. Samples likely to contain any species and unlikely to contain an already identified species are selected based on a high value of $y_{any}^* \cdot (1 - y_{spec,max}^*)$, where $y_{spec,max}^*$ denotes the maximum value across all species.

In total, combining two layer modes (fine-tuning, frozen) with three diversity-based selection strategies (Core-Set, HDBSCAN, K-means) across four embedding dimensionalities (2, 16, 128, 1024), together with two additional uncertainty-based selection strategies (Ratio-max, Dual-Classifier) that use the fixed 1024-dimensional embedding, yields 28 distinct active learning configurations.

Variational autoencoder methods

Within self-supervised learning methods, we evaluate variational autoencoder-based methods with an encoder of three convolutional layers (16, 32 and 64 filters) and a decoder that mirrors this structure. The shallow architecture balances representation capacity and computational efficiency while capturing hierarchical spectrogram features. Inputs are 224x 224 pixel spectrograms computed with scikit-maad (Ulloa et al., 2021). Variational autoencoders are trained exclusively on the current labelled dataset. Dimensionality is varied (2, 16, 128, 1024) via the encoder-decoder interface to balance representation richness and generalization, with lower dimensions supporting clustering and outlier detection, and higher dimensions capturing finer detail (Bengio et al., 2013). We test two variational autoencoder methods:

1. VAE-Reconstruction minimizes the reconstruction error, selecting unlabelled samples with the highest errors.
2. VAE-Classification trains the encoder with a shallow classification head alongside the decoder. All samples are encoded, and unlabelled samples farthest from the nearest labelled sample are selected, following our results showing Core-Set performs best among active learning methods (Section 3).

In total, combining four embedding dimensionalities (2, 16, 128, 1024) and two selection strategies (Reconstruction, Classification) yields 8 distinct variational autoencoder configurations.

Similarity learning methods

Inspired by Zhu et al. (2018), two audio samples are embedded using BirdNET, and their embeddings—along with their absolute difference—are concatenated into a 3072-dimensional feature vector. This is passed through two fully connected layers with 256 and 64 nodes, followed by a single-node output layer that predicts a similarity score. Models are trained on the current labelled dataset. We compare embeddings from the frozen BirdNET model with those obtained after fine-tuning on the current labelled dataset. All subsequent layers are fine-tuned in both cases. With n_t^l samples in the current labelled dataset, the number of distinct sample pairs grows as $O(n_t^{l^2})$. To maintain computational feasibility, we randomly sample up to 10,000 pairs when fine-tuning BirdNET, and up to 100,000 pairs when using its frozen embeddings. We compare weak similarity with strong similarity. For sample selection, each unlabelled sample is paired with each labelled sample. For each unlabelled sample, the maximum similarity to any labelled sample is computed, and samples with the lowest maximum similarity are selected. In total, combining two layer modes (fine-tuning, frozen), and two similarity definitions (weak, strong) yields 4 distinct similarity learning configurations.

Audio-language model methods

Prompt responses are generated using the audio-language model NatureLM-Audio. We employ three distinct configurations, each comprising a prompting stage to generate model outputs in JSON format and a subsequent decision protocol that selects the next samples to query. The following outlines the core principles and decision process of these methods, with the full prompts provided in Appendix B.

1. ALM-Classification instructs the model directly with the names of detected, dataset-relevant species. If a species cannot be identified, the model is prompted to provide a short descriptive label, such as 'bird, rattling'.
2. ALM-Taxa-OneStep instructs the model to detect dataset-relevant animal vocalizations in an audio recording and classify each along the taxonomic hierarchy. The model predicts taxonomic ranks from domain to species, reporting each level only when reliably identified.
3. ALM-Taxa-TwoStep follows the same principle as ALM-Taxa-OneStep but splits the process into two sequential prompts. The first prompt identifies samples containing dataset-relevant animal vocalizations, which are then passed to a second prompt that classifies the detected vocalizations using the same hierarchical reasoning as ALM-Taxa-OneStep.

The query process iteratively selects samples containing detected dataset-relevant animals, prioritizing those predicted to include the

most previously unobserved taxonomic entities—defined as species or descriptive labels in ALM-Classification, and as unique taxonomic combinations in ALM-Taxa-OneStep and ALM-Taxa-TwoStep.

2.4.3 | Experimental set-up

As the annotations in the datasets do not cover all present animals but focus on specific taxonomic groups (Table 2), we use the annotated species-level labels as the target class set to be identified, reflecting scenarios in which only specific taxa are of interest.

At the initial iteration ($t = 1$), all samples are unlabelled. We use a batch size of 10 samples per iteration and a stop budget of 500 samples. Sample selection follows the query method configurations described in Section 2.4.2. For methods requiring labelled samples (e.g. to train a model), the first 10 samples are selected randomly.

Each model is trained for 50 epochs with a learning rate of 10^{-4} . For classification models, binary cross-entropy is used for multi-label data, and categorical cross-entropy is applied for single- or multi-class data, such as when predicting the presence or absence of any species.

The query methods are evaluated using the framework and metrics in Section 2.2, with each experiment repeated over five independent runs.

The computational analyses, including pre-processing, are conducted in Python 3.11. We use BirdNET V2.4, referred to as BirdNET throughout this paper. All experiments are run on a system with an NVIDIA A100 Tensor Core GPU (80 GB) and two CPU cores.

3 | RESULTS

To assess the proposed framework (Section 2.2), we apply it to all combinations of query method configurations and datasets described in Section 2.4, and report both qualitative and quantitative results.

3.1 | Qualitative results

Figure 3 shows qualitative results for a representative selection of query methods across all datasets. The complete set of qualitative outcomes is provided in Appendix C.

General trends in overall performance across query method configurations and datasets, as shown in the top row of Figure 3, indicate that learning curves rise steeply in the early iterations, meaning more classes are identified at the beginning. Class-specific performance, shown in the middle row of Figure 3, shows that more frequent classes in the dataset are generally identified earlier, whereas less frequent classes appear later or remain unidentified. Regarding computational cost, shown in the bottom row of Figure 3, processing time varies substantially (note the logarithmic scale).

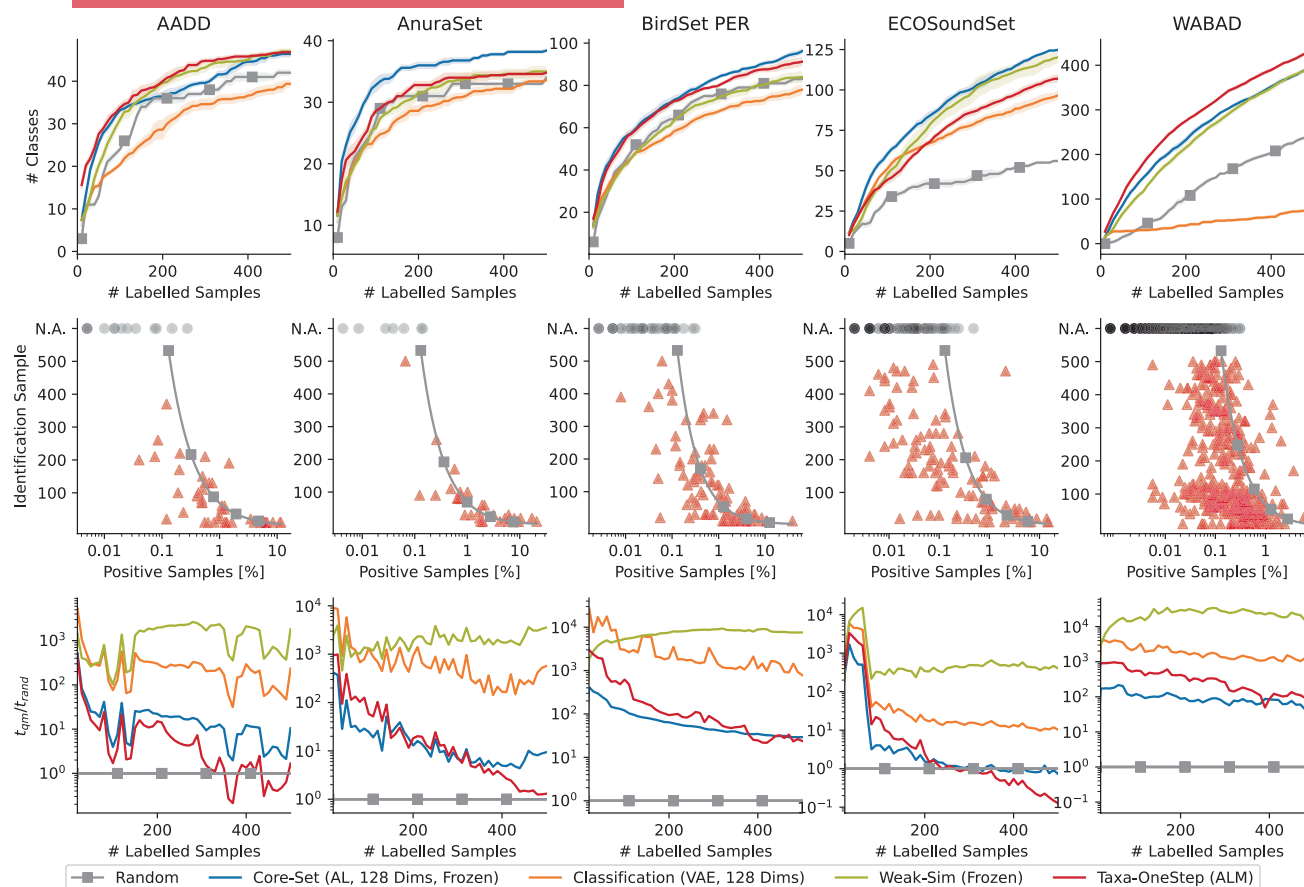


FIGURE 3 Selection of qualitative results. The top row shows overall performance, the middle row shows class-presence-specific performance for ALM Taxa-OneStep (classes identified within the 500-sample budget are marked with red triangles [Δ], others with black circles [$\bullet$]), and the bottom row shows processing time. Representative query methods from different method families are shown: Core-Set (active learning, 128-dimensional, frozen), Classification (variational autoencoder, 128-dimensional), Weak Similarity (frozen), and Taxa-OneStep (audio-language model). For comparison, all figures include the random baseline, shown as a grey line with grey squares ($\square$).

3.2 | Quantitative results

While qualitative results provide illustrative examples, we focus on quantitative analysis to examine the effects of embedding dimensionality, layer mode and query method selection. Representative quantitative results are presented below for clarity and ease of interpretation, with the complete set of results provided in [Appendix D](#).

3.2.1 | Impact of embedding dimensionality

To investigate the impact of embedding dimensionality, [Figure 4](#) presents quantitative results for all query methods operating in embedding spaces—including Core-Set, HDBSCAN, K-means, and VAE-based approaches—averaged over the five datasets. Individual dataset results are provided in Section D.1 in [Appendix D](#). Results indicate that overall performance (see [Figure 4](#), left) remains largely constant across embedding dimensionalities, with the exception of the Core-Set query method, which shows improved performance in higher-dimensional

spaces. Class-presence-specific performance (see [Figure 4](#), centre) indicates that, with increasing dimensionality, Core-Set improves, peaking at 128 dimensions, while HDBSCAN and K-means degrade, and VAE-based methods remain largely constant. Processing time (see [Figure 4](#), right) remains in the same order of magnitude and thus appears approximately constant on the logarithmic scale when fine-tuning, as fine-tuning dominates the computational cost compared to processing higher dimensional embeddings. In contrast, for frozen embeddings, processing time increases noticeably with dimensionality.

3.2.2 | Effect of layer mode (frozen vs. fine-tuned)

To investigate the impact of layer mode, [Figure 5](#) presents quantitative results for all query methods operating in embedding spaces that can be either frozen or fine-tuned, including active learning and similarity-based methods, averaged over the five datasets. As 128-dimensional embeddings show slight performance improvements, this dimensionality is used whenever a query method depends on an embedding dimensionality. Results indicate that overall performance (see [Figure 5](#),

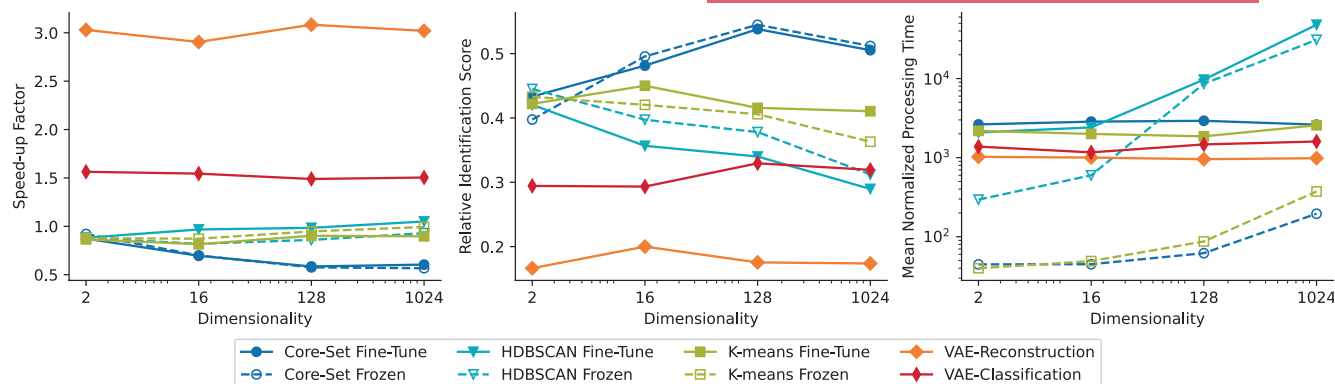


FIGURE 4 Quantitative results for all query methods evaluated across embedding dimensionalities (2, 16, 128, and 1024), aggregated over datasets. Left: Speed-up factor (overall performance, lower is better); middle: Relative identification score (class-specific performance, higher is better); right: Mean normalized processing time (lower is better).

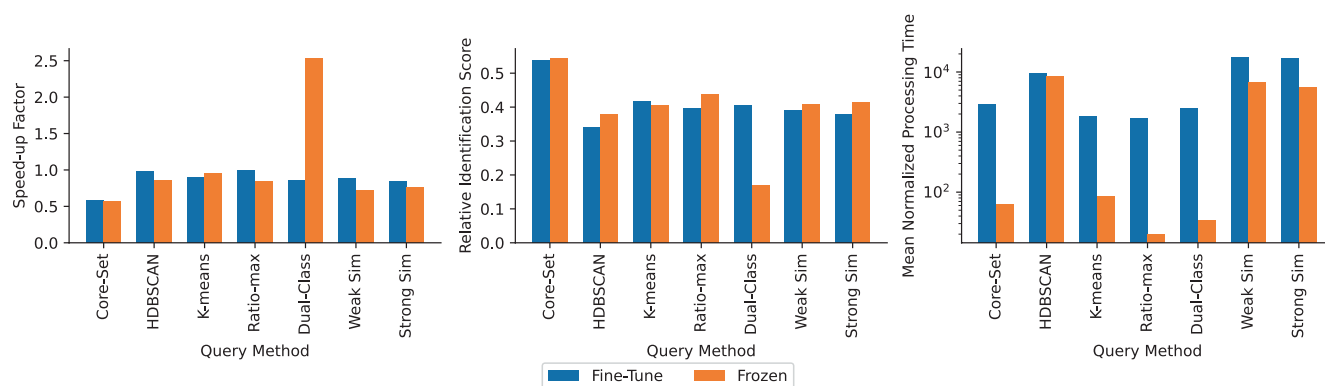


FIGURE 5 Quantitative results for all query methods evaluated across layer modes (fine-tuned, frozen), aggregated over datasets. For methods that depend on embedding dimensionality, a dimension of 128 is used. Left: Speed-up factor (overall performance, lower is better); middle: Relative identification score (class-specific performance, higher is better); right: Mean normalized processing time (lower is better).

left) and class-presence-specific performance (see Figure 5, center) are generally higher when using frozen embeddings for most query methods, as reflected by lower speed-up factor values and higher RIS values. Exceptions include K-means, which shows slightly improved performance with fine-tuned embeddings, and Dual-Class, which exhibits a substantial drop in performance when using frozen embeddings. Processing time (see Figure 5, right) is consistently higher when using a fine-tuned embedding space.

3.2.3 | Comparison of query methods

To investigate query method effectiveness, Table 3 presents quantitative results for all query methods, averaged over the five datasets. As a dimensionality of 128 and the frozen layer mode show slight performance improvements, these settings are used whenever a query method depends on them.

The additional baseline, Random CNN, shows performance similar to that of the random sampling baseline. The following presents the results of Table 3, organized by method family. These findings align with the qualitative results shown in Figure 3 and Appendix C.

Active learning methods

Table 3 shows that Core-Set exhibits the best overall and class-specific performance across query methods, significantly outperforming random sampling. HDBSCAN, K-means, and Ratio-max show slight improvements over random sampling, while Dual-Class exhibits substantially decreased performance. Processing time analysis shows that HDBSCAN requires approximately three orders of magnitude more time than random sampling, while the other query methods are one order of magnitude higher.

Variational autoencoder methods

Table 3 shows that the overall and class-specific performance of the Reconstruction query method is substantially below that of random sampling. The Classification query method shows some improvement, but still does not outperform random sampling. Processing time analysis shows that both methods require approximately three orders of magnitude more time than random sampling.

Similarity learning methods

Table 3 shows that the overall and class-specific performance of both similarity definitions significantly outperforms random sampling.

Method fam.	QM	Speed-up factor (mean $\pm$ SEM)	RIS (mean $\pm$ SEM)	Mean normalized processing time (mean $\pm$ SEM)
Baseline	Random	1.00 $\pm$ 0.00	0.31 $\pm$ 0.09	1 $\pm$ 0
	Random CNN	1.20 $\pm$ 0.18	0.31 $\pm$ 0.11	3 $\pm$ 1
Active Learning	Core-Set	0.58 $\pm$ 0.09	0.54 $\pm$ 0.06	62 $\pm$ 14
	HDBSCAN	0.86 $\pm$ 0.07	0.38 $\pm$ 0.08	8581 $\pm$ 5508
	K-means	0.95 $\pm$ 0.10	0.41 $\pm$ 0.08	87 $\pm$ 40
	Ratio-max	0.85 $\pm$ 0.06	0.44 $\pm$ 0.05	20 $\pm$ 6
	Dual-Classifer	2.54 $\pm$ 0.49	0.17 $\pm$ 0.13	35 $\pm$ 8
VAE	Reconstruction	3.08 $\pm$ 0.83	0.18 $\pm$ 0.13	956 $\pm$ 283
	Classification	1.49 $\pm$ 0.47	0.33 $\pm$ 0.09	1469 $\pm$ 618
Similarity	Weak-Sim	0.72 $\pm$ 0.12	0.41 $\pm$ 0.08	6790 $\pm$ 4025
	Strong-Sim	0.77 $\pm$ 0.10	0.42 $\pm$ 0.07	5675 $\pm$ 2953
ALM	Classification	0.68 $\pm$ 0.10	0.49 $\pm$ 0.06	682 $\pm$ 318
	Taxa-OneStep	0.64 $\pm$ 0.09	0.53 $\pm$ 0.06	233 $\pm$ 73
	Taxa-TwoStep	0.64 $\pm$ 0.09	0.51 $\pm$ 0.07	180 $\pm$ 56

Note: Bold values indicate the best performance for each metric among all compared methods.

TABLE 4 Experimental results of audio-language model methods. The table reports positive annotations (percentage of samples containing at least one annotated target species sound) and the number of taxonomic entities. Results are shown for manual annotations that are considered ground-truth (GT), and for the audio-language model methods ALM-Classification (CL), ALM-Taxa-OneStep (T1), and ALM-Taxa-TwoStep (T2). For taxonomic entities, the number of identified entities (Found) is reported for all methods. For ALM-Classification, the numbers of found scientific names (Sci. Names) and descriptive names (Desc. Name) are presented. For ALM-Taxa-OneStep and ALM-Taxa-TwoStep, the numbers of entities with unknown species but higher-level taxa (Sp. unk.) and invalid taxonomic hierarchies (Inv. Tax.) are reported.

Dataset	Positive annotations (%)				# taxonomic entities									
	GT				GT	CL			T1			T2		
		CL	T1	T2		Found	Sci. Name	Desc. Name	Found	Sp. unk.	Inv. Tax.	Found	Sp. unk.	Inv. Tax.
AADD	63.7	99.1	100.0	98.2	58	572	563	9	522	24	241	252	3	108
AnuraSet	62.0	92.6	99.9	75.2	42	1007	809	198	765	5	565	765	18	349
BirdSet PER	86.6	97.2	100.0	98.1	130	1249	1220	29	900	4	388	686	4	297
ECOSoundSet	92.6	90.1	100.0	50.3	200	534	464	70	597	36	381	259	7	102
WABAD	77.8	93.2	99.6	89.5	1116	3690	3587	103	3596	12	1551	2657	17	1103

Processing time analysis shows that both methods require approximately three orders of magnitude more time than random sampling.

Audio-language model methods

Table 3 shows that the overall and class-specific performance of all audio-language model query methods significantly outperforms random sampling, achieving the best or nearly the best results across all query methods. Processing time analysis shows that audio-language model methods require approximately two orders of magnitude more time than random sampling.

Presenting dataset-specific details, Table 4 reports, for each prompt and dataset, the proportion of samples predicted to contain target species, along with the corresponding predicted number of

taxonomic entities. Predicted target species presence exceeds the fraction of samples containing target species (ground truth) for all datasets except ECOSoundSet. The predicted number of taxonomic entities consistently exceeds the number of target species (ground truth), with ALM-Classification predicting the highest, followed by ALM-Taxa-OneStep and ALM-Taxa-TwoStep. Using the `species` module from the Python library `pygbif` (<https://pygbif.readthedocs.io/en/latest/modules/species.html>), the number of scientific and descriptive names is derived for ALM-Classification, and the number of invalid taxonomic hierarchies is derived for ALM-Taxa-OneStep and ALM-Taxa-TwoStep. A taxonomic hierarchy is considered invalid if an 'unknown' rank is followed by a specified lower rank, or if any specified rank is inconsistent with its designated parent. Table 4

TABLE 3 Quantitative experimental results sorted by method family (Method Fam.) and query method (QM). Displayed are the speed-up factor (lower is better), the relative identification score (RIS; higher is better), and the mean normalized processing time (lower is better). For methods that depend on embedding dimensionality, a dimension of 128 is used. For methods that depend on layer mode, the frozen mode is used. Results show the mean and standard error of the mean (SEM) across datasets.

shows that ALM-Classification predicts considerably more scientific names than descriptive names. For both hierarchical methods, approximately 50% of the identified taxonomic hierarchies are invalid across all datasets, while fewer than 10% of species are labelled as unknown.

4 | DISCUSSION

To guide future work on NCI, we interpret the qualitative and quantitative findings across all datasets and query method configurations.

4.1 | Impact of embedding dimensionality

Previous studies evaluating embedding spaces from pre-trained bioacoustic models have shown that reducing embeddings to lower dimensional spaces can improve performance. For instance, Kather et al. (2025) report that both classification and clustering performance on single-label datasets improve when the embedding dimensionality is reduced to 300 using UMAP for dimensionality reduction. Kath et al. (2024a) report improved species identification results on multi-label datasets when reducing the dimensionality of BirdNET embeddings to 10 by adding and training an additional embedding layer using the labelled dataset. In contrast to these findings, our results show no clear trend indicating that reducing dimensionality enhances NCI performance. While HDBSCAN and K-means tend to perform better at lower dimensionalities, Core-Set shows improved performance in higher dimensional spaces, and VAE-based methods exhibit no clear correlation with dimensionality. These results indicate that BirdNET's embedding space effectively groups species and that high-dimensional embeddings retain relevant species information, as evidenced by Core-Set's improved performance, but it is challenging to cluster since query methods based on the clustering algorithms HDBSCAN and K-means perform better at lower dimensionalities.

4.2 | Effect of layer mode (frozen vs. fine-tuned)

While prior work in bioacoustic classification tasks indicates that fine-tuning BirdNET improves performance (Section 2.3.1), our results show that a frozen embedding space generally enhances NCI performance compared to fine-tuning. The only exception is the Dual-Classifier query method, whose performance is significantly below random when using frozen embeddings. The improved performance of frozen embeddings is likely due to the differing objectives of classification and NCI. Fine-tuning substantially reorganizes embedding spaces by increasing separation of task-relevant classes and improving their separability in the learned representation space (Kumar et al., 2022; Rajaei & Pilehvar, 2021). Unlike classification, where optimizing separation of known

classes is beneficial, NCI aims to identify previously unseen classes. Consequently, fine-tuning biases the embedding space toward known classes, potentially reducing the representational capacity needed to distinguish novel classes and hindering their identification.

4.3 | Comparison of query methods

4.3.1 | Active learning methods

Although active learning query methods are designed to accelerate classification improvement, our results show that they also outperform random sampling in identifying new classes. For diversity-based methods, this is likely because the embedding space effectively groups samples according to species presence (Ghani et al., 2023; Kath et al., 2024a). For uncertainty-based methods, such as Ratio-max, targeting uncertain samples often selects borderline cases that include novel species, thereby improving the identification of new classes compared to random sampling. Dual-Classifier's scoring function and its reliance on classifiers trained on known species likely bias selection toward familiar regions, limiting exploration of novel-class embeddings.

4.3.2 | Variational autoencoder methods

When using the current labelled dataset for training and selecting samples from the unlabelled dataset based on the highest reconstruction error, we find that semantic shifts are poorly detected, with performance significantly below random. This result is in contrast to the findings of Ntalampiras and Potamitis (2021), who improve species identification using this method. Adjusting the encoding function by adding a shallow classifier that operates on the encoder's output and sampling within the resulting latent space improves performance, although it remains below that of random sampling. The low performance is likely due to the iterative nature of NCI, which results in comparatively small labelled datasets at each iteration and thus provides only limited representations of the species present. In contrast to the setup used by Ntalampiras and Potamitis (2021), this constrains the model's ability to learn a function that reliably captures the semantics of the labelled dataset.

4.3.3 | Similarity learning methods

Comparing weak to strong similarity reveals no clear performance difference; both outperform random sampling. Using frozen BirdNET weights, however, yields slightly better overall and class-specific performance. This is likely due to the limited number of labelled samples in each iteration, which causes individual samples to appear in multiple similarity pairs and increases the risk of overfitting (Yosinski

et al., 2014). Frozen weights are unaffected by this effect, explaining their marginally better performance.

4.3.4 | Audio-language model methods

Across all prompts, results show a clear and consistent improvement over random sampling. However, the results also reveal an issue of overconfidence. Almost all samples are predicted to contain target species, and when asked to provide the scientific name if confident or a descriptive name if not, the model outputs predominantly scientific names—far exceeding the number of target species actually present in the datasets. Targeting this weakness by asking the model to output a complete taxonomic hierarchy and to explicitly label ranks as unknown when uncertain reduces the number of predicted taxonomic entities. However, the model still produces far more entities than are actually present, rarely uses the unknown label, and yields approximately 50% invalid taxonomic hierarchies. This behaviour is likely a consequence of the NatureLM-Audio training process, which encourages assigning species names to nearly any sound but does not explicitly model hierarchical taxonomic structure, resulting in no mechanism to constrain predictions to the queried target species (Robinson et al., 2025). Solutions to mitigate this type of hallucination include estimating model certainty, for example by leveraging the model's internal entropy (Farquhar et al., 2024), or by training an audio-language model using a hierarchical approach. Such hierarchical training has been applied in the bioimage domain, as demonstrated by Bjerger et al. (2023) for insect classification using convolutional neural networks and by Stevens et al. (2024) for plant classification using transformer-based models.

5 | CONCLUSION

To support global biodiversity monitoring efforts that depend on efficient species identification in large-scale passive acoustic monitoring (PAM) datasets, this work introduces the machine learning task of novel class identification (NCI), which aims to maximize the number of identified classes (e.g. species or call types) while minimizing the number of samples examined. NCI is formally defined and positioned within related machine learning tasks, with a comprehensive evaluation framework proposed to assess overall and class-presence-specific performance through qualitative visualizations and quantitative metrics. Based on insights derived from the preceding literature analysis, the four promising query method families of embedding-based learning, self-supervised learning, similarity learning and audio-language models are identified as particularly relevant for addressing the NCI task. To inform and guide future research, initial experiments systematically explore query methods across these method families, varying embedding dimensionality (2, 16, 128 and 1024) and layer mode (fine-tuning vs. freezing pre-trained weights) configurations where applicable. In total, 44 configurations are evaluated across five real-world multi-label PAM datasets.

Our initial experiments show that dimensionality has little consistent impact, that using frozen embeddings yields slightly better performance than fine-tuning the model, and that Core-Set and all methods from the audio-language model family achieve the highest query performance.

Our findings suggest three directions for future work in NCI. First, across all query methods, classes with higher representation are generally identified earlier than less frequent classes, suggesting that an effective NCI query method should prioritize the identification of sparse classes. Since PAM datasets contain a substantial fraction of samples without target species sounds (20%–40% in the datasets used here), filtering out recordings without relevant biological activity would significantly reduce the search space for new target species. The Dual-Classifier and audio-language model methods attempt to exclude such samples, but with limited success (see Table 4 for audio-language model results). Another challenge arises with multi-label data: combinations of already identified species can appear anomalous, and species that occur only alongside others are difficult to distinguish. Embedding-based methods may struggle in these cases, as co-occurring known species can obscure novel or rare species within the embedding space. Separating sound sources to create single-label data could help mitigate this issue. Approaches such as BamBird (Michaud et al., 2023) or Bird-MixIT (Denton et al., 2022) appear promising for pre-filtering, but integrating them is beyond the scope of this study. Second, while this work assumes zero knowledge about the target classes, practical settings may include prior knowledge. This is particularly useful when target species are included in the training data of a classification model. For example, (Kath, Serafini, et al., 2024; Kath, Troshani, et al., 2024) aim to identify as many species (including birds) as possible from PAM data, and use BirdNET to identify species already present in its training data, providing an initial set of identifications that helps guide the NCI model towards the underlying structure of the target class set. Third, audio-language models show strong potential, and their performance could be further improved by explicitly training them on hierarchical taxonomic structures. In addition, instead of pre-computing prompt outputs and sampling from these, audio-language models could be integrated directly into the iterative NCI process—using the currently labelled samples and identified species as context and querying the unlabelled dataset for samples likely containing previously unseen classes. Implementing such an approach, however, requires a substantial reduction in audio-language model processing time. It was therefore not included in this study but remains an important direction for future work.

Key open questions in NCI include when to stop sampling—that is, how to determine that all classes have been identified—and how query size impacts performance. Smaller query sizes may improve performance but typically increase computational cost, as most query methods scale poorly with the number of iterations. Previous work (Kath et al., 2024a) considered a query size of 1; however, for many sampling strategies, this is computationally infeasible, highlighting the need to balance accuracy and efficiency. One limitation of this study on PAM data is that we evaluate query methods

only at the species level as the target class definition, while performance on more fine-grained annotations such as call types and more coarse-grained soundscape categories is not investigated. Another limitation is the restricted set of target classes to dataset-specific annotated taxa (Table 2), excluding species outside each dataset's scope.

AUTHOR CONTRIBUTIONS

Hannes Kath conceived the ideas, designed the methodology, performed the analyses and led the writing of the manuscript. Thiago S. Gouvêa supervised the research project and revised the manuscript. Daniel Sonntag provided guidance during the research process and contributed to reviewing the final manuscript. All authors contributed to the drafting process and approved the final version for publication.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210x.70373>.

DATA AVAILABILITY STATEMENT

All code and data used in this study are publicly available. The main experiments for Novel Class Identification can be accessed at <https://github.com/HKathman/Novel-Class-Identification>, while preprocessing for audio-language-model-based methods using NatureLM-Audio is provided at <https://github.com/HKathman/NatureLM-Audio>. A versioned archive of the release is available via Zenodo: <https://doi.org/10.5281/zenodo.21143210> (Kath, 2026). All datasets used in this study have been published previously and are accessible via the references provided in the manuscript; no new data were generated.

DATA SOURCES

Data sources include the publicly available datasets as described previously (Cañas et al., 2023; Funosas et al., 2026; Pérez-Granados et al., 2026; Rauch et al., 2025; Weldy et al., 2024). Full citations are provided in the reference list.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix A. Relative identification score.

Appendix B. Audio-language models prompts.

Appendix C. Comprehensive qualitative results.

Appendix D. Comprehensive quantitative results.

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